

Muhammad Zuhaib

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Education

Bachelor of Engineering in Computer Systems Engineering

2022–2026

Sukkur IBA University, Sukkur, Sindh, Pakistan

- CGPA: 3.67 / 4.00 | Rank: Top 3 in cohort
- **Bachelor's Thesis:** “Integrating Classical and Quantum Models for a Hybrid Quantum Classical Machine Learning Framework for Lung Disease Classification” (Supervisor: Dr. Junaid Ahmed)
- Developed an automated pipeline (Input: Chest X-ray image → Quantum encoding → Quantum Variational Circuit (QVC) → Measured in Classical State → State-of-the-Art Models (ViT, ResNet and CNN) → Classification) for five-class lung disease classification including Bacterial Pneumonia, Viral Pneumonia, COVID-19, Tuberculosis, and Normal lung conditions.

Research Interests

Computer Vision | Deep Learning | Quantum Machine Learning | Structural Health Monitoring | Signal Processing | Image Processing | Intelligent Sensing Systems | Autonomous Multi-Agent Systems | Agentic AI Chatbots

Technical Skills

Programming & Databases: Python, C#, MATLAB, SQL, MySQL

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LLMs, SHAP

Computer Vision & Signal Processing: OpenCV, YOLO, ViT, Image Processing

Frameworks & Tools: ASP.NET, Streamlit, Flask, n8n, Edge AI (TF Lite, ESP32), Git, Linux

Publications

Accepted and Published

- Muhammad Zuhaib, M.Faizan, A.Rehman, J. Ahmed, “Evaluating Deep Learning and Hybrid Quantum Algorithms for Lung Disease Classification.” *iCoMET 2026*, IEEE Xplore, 2026.

Currently Working On

- Muhammad Zuhaib, M.Faizan, A.Rehman, J. Ahmed. “Quantum-Enhanced Chest X-ray Disease Detection Through Patch-Based Feature Encoding and Hybrid Learning.”

Conference Presentations

- International Conference on Computing, Mathematics & Engineering Technologies (iCoMET 2026) – “Evaluating Deep Learning and Hybrid Quantum Algorithms for Lung Disease Classification.” (Paper Presentation). Sukkur IBA University, Pakistan. May 2026. DOI: 10.1109/iCoMET69771.2026.11591621

Research & Development Projects

Book Inventory AI Agent [Code]

2026

- Built an n8n-based autonomous agent using Groq LLaMA 3 and Airtable to manage book inventory through natural-language commands, automating lookup, updates, and stock tracking.

Website Status Monitor Agent [Code]

2026

- Developed an n8n AI agent that continuously monitors website uptime and health, triggering automated alerts and status reports without manual oversight.

AI Emotion Detection System [Code]

2025

- Built an NLP-powered classifier detecting six emotions (Happiness, Sadness, Anger, Love, Hate, Surprise) using TF-IDF, keyword boosting, VADER sentiment features, and an XGBoost classifier trained on a balanced, oversampled dataset.
- Designed an interactive GUI with confidence bars and emotion-distribution visualizations for real-time text analysis.

AI Voice Cloning Studio [Code]

2025

- Developed a multilingual voice cloning application using Coqui XTTS v2, supporting live microphone recording, noise reduction, and zero-shot voice synthesis across six languages.
- Implemented audio preprocessing (mono conversion, normalization, high-pass filtering) with an interactive interface built on ipywidgets.

AI-Powered Resume Analyzer and Job Matching [Code] 2025

- Built an NLP pipeline extracting skills from resume PDFs and matching candidates to relevant job postings via live external job APIs.
- Added auto-generated professional summaries, visual skill analysis charts, and personalized career guidance recommendations.

AI Content Marketing Automation System [Code] 2025

- Built a Google Sheets sidebar add-on automating blog content generation using the OpenAI GPT API paired with the Unsplash API for free image sourcing.
- Designed a dark-mode sidebar UI in Google Apps Script for a fully automated, end-to-end content workflow.

Real-Time Speech Emotion Recognition using LSTM and MFCC Features [Code] 2025

- Trained an LSTM classifier on MFCC audio features for real-time emotion classification from speech signals.

Smart Sign Glove [Code] 2025

- Designed a low-cost smart glove that translates sign language into text using flex sensors to detect finger movements, recognizing both individual letters and common words with real-time feedback.

JPEG and Wavelet-Based Image Compression [Code] 2024

- Implemented and compared JPEG (DCT-based) and wavelet-based compression techniques, analyzing compression ratio versus image quality trade-offs.

Leaf Disease Detection [Code] 2024

- Built a CNN-based classifier to detect and categorize plant leaf diseases from image data for early agricultural diagnosis.

House Price Prediction Using Linear Regression [Code] 2024

- Developed a regression pipeline with feature engineering to predict housing prices from structured tabular data.

SmartOffice Model Using Cisco Packet Tracer [Code] 2024

- Simulated a smart office network integrating IoT devices, smart security systems, and network management for secure, automated office infrastructure.

Experience

Machine Learning Intern | Ezitech Institute, Rawalpindi june 2025–Aug 2025

- Completed an 8-month internship at Ezitech Institute, a Rawalpindi-based technology training institute, working on applied machine learning projects as part of its Artificial Intelligence internship track.
- Gained hands-on, industry-oriented experience in model development, data preprocessing, and deployment workflows under structured mentorship.
- [View Certificate](#)

Generative AI Co-Trainer | Department of Computer Systems Engineering, Sukkur IBA University Month 2025–Month 2026

- Co-facilitated training sessions on Generative AI concepts and applications for students in the Department of Computer Systems Engineering.
- Designed hands-on exercises and practical demonstrations to help students build applied skills in generative AI tools and workflows.
- [View Certificate](#)

Honours and Awards

- **Prime Minister's Laptop Award** (2026) – National merit-based award for academic excellence, Government of Pakistan.
- **Sindh Educational Endowment Fund (SEEF) Scholarship** (2022–2026) – Fully merit-based undergraduate scholarship, Government of Sindh.

Specializations & Certifications

- Machine Learning Specialization – DeepLearning.AI / Coursera – View Certificate
- Mathematics for Machine Learning and Data Science – DeepLearning.AI / Coursera
- Deep Learning Specialization – DeepLearning.AI / Coursera – View Certificate
- foundation of deep learning and Neural Networks – Packt/Coursera– View Certificate
- Large Language Models – Udemy
- Google Prompting Essentials Specialization – Google / Coursera – View Certificate
- Google IT Support – Google / Coursera – View Certificate

Volunteer Work

- Executive Member** | Computer Systems Engineering Society 2023-2024
- Served as an executive member of the Computer Systems Engineering Society, contributing to society activities and initiatives.
- Project Exhibition Participant** | Sukkur IBA University 2025
- Participated in the university’s project exhibition and secured **1st Position** among competing teams.
- Project Exhibition Participant** | IBA Community College, Naushahro Feroze 2021
- Participated in the project exhibition held at IBA Community College, Naushahro Feroze.
- Computer Engineering Workshop** | Project Exhibition 2022
- Participated in the project exhibition conducted as part of a Computer Engineering workshop.

References

Dr. Junaid Ahmed Bhatti
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